Hong Kong Marine Water Quality Analysis

Background

Hong Kong's marine environment is vital to its ecological and economic systems, with over 1,650 kilometers of coastline supporting a diverse array of marine species. However, rapid urbanization and high population density have led to environmental pressures. The Hong Kong Environmental Protection Department (EPD) monitors marine water quality by tracking numerous indicators. The government has been working to improve marine conditions. However, it remains unclear if these measures have significantly improved marine conditions.

Research Focus

This study assesses two significant key indicators of water quality—dissolved oxygen (DO) and biochemical oxygen demand (BOD_5)—between 2020 and 2024. We aim to determine if observed changes are true improvements or just natural variations. The findings will offer insights into the change in the marine environment and future environmental policies.

The dataset for this study was extracted from data.gov.hk. It includes water samples collected from various monitoring stations across Hong Kong's marine zones, with measurements taken at different depths (surface, middle, and bottom) from 2020 to 2024.

Specifically, this study focuses on the following three parameters:

Parameter	**Description**	**Interpretation**
Dissolved Oxygen (DO)	Amount of oxygen available for aquatic life (mg/L)	Higher DO indicates better water quality
5-day Biochemical Oxygen Demand (BOD_5)	The amount of oxygen consumed by microorganisms that decompose organic matter	Higher BOD_5 indicates worse water quality

These three parameters were selected because they collectively represent the core dimensions of marine ecological health and organic pollution.

- Dissolved Oxygen (DO) directly influences the survival of aquatic organisms and serves as a fundamental measure of a water body's capacity to sustain life. Low DO levels often indicate excessive organic pollution or eutrophication.

- Biochemical Oxygen Demand (BOD_5) measures the oxygen consumed during microbial decomposition of organic matter.

Together, these indicators provide a comprehensive picture of oxygen balance, organic pollution, and particulate matter concentration, which are essential factors for evaluating the ecological condition of Hong Kong's marine waters.

In this study, we will conduct independent t-tests to compare the mean values of each water quality indicator between 2020 and 2024, determining whether changes are statistically significant, and perform regression analysis to explore how factors such as region, depth, and sampling time may influence overall water quality trends.

Methodology

1. Data Import

Obtain a twenty-five-year dataset (2001–2024) from the Environmental Protection Department (EPD) in Hong Kong. In the hypothesis section, we only use the 2020 and 2024 datasets; for the regression section, the past 24 years' dataset will be used..

2. Data Cleaning

To ensure consistency and comparability, we apply the following steps:

1. **Grouped Water samples based on depths:** Water samples collected from different depths will be analyzed separately (Surface, Middle, and Deep Water).
2. **Selected key indicators:** Dissolved Oxygen (mg/L) and 5-day Biochemical Oxygen Demand (BOD_5, mg/L)
3. **Removed missing or invalid values**, including symbols like <0.5.
4. **Converted all measurements to a numeric format** for analysis.

This ensures a clean dataset suitable for statistical testing.

3. Statistical Method

We test whether there is a significant difference in the mean values of each indicator between **2020** and **2024**:

Parameter	Null Hypothesis (H_0)	Alternative Hypothesis (H_1)	Expected Direction
Dissolved Oxygen (DO)	$\mu_{(2020)} = \mu_{(2024)}$	$\mu_{(2020)} < \mu_{(2024)}$	Higher DO is better

BOD_5	$\mu_{(2020)} = \mu_{(2024)}$	$\mu_{(2020)} > \mu_{(2024)}$	Lower BOD_5 is better

The **significance level** (α) is set at **0.05**. If $p < 0.05$, we reject the null hypothesis (H_0), indicating a significant difference. A t-test in Python helps determine the p-value, showing the chance that the observed difference is due to random variation. The t-statistic, which compares the sample mean to the population mean, is calculated by dividing the difference by the standard error; a larger value suggests stronger evidence against H_0. Type errors are also important. The Type I error occurs when we incorrectly reject H_0, while the Type II error reflects the failure to reject a false H_0. Using the statsmodels module in Python, we first measure the power of the test and then calculate the Type II error rate (β).

Results

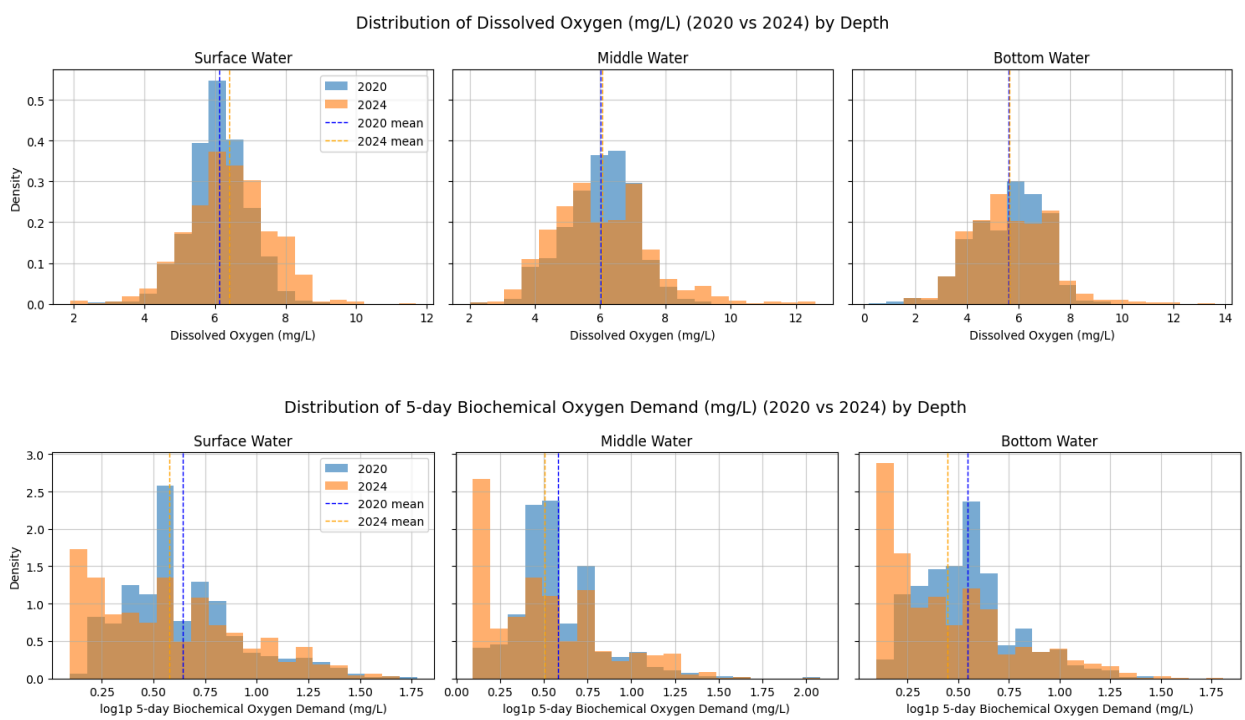

Fig. 1. Distribution of DO and BOD_5 levels by depth in 2020 and 2024

To ensure a comprehensive analysis, we conduct studies and obtain results at three different depths (please note that the specific depths are not provided in the data, but an official classification of surface, middle, and bottom water is). Figure 1 shows that, in all DO distributions, the mean line for 2024 is closer to the right of the mean for 2020, indicating higher overall dissolved oxygen levels. For the BOD_5 distribution, however, the 2024 mean is smaller than the 2020 mean, suggesting that the marine environment is under less organic load.

Fig. 2 Average DO and BOD_5 by water zones in 2020 and 2024

Assessments from Figure 2 indicate that DO levels have improved at all depths in 2024 for the surface and bottom water levels, with lower BOD_5 in most areas, particularly Victoria Harbour and Deep Bay. This reflects an overall improvement in water quality. The p-value and t-statistic results based on the mean difference are hence calculated:

DO	Surface Level	Middle Level	Bottom Level
p-value	$<1\times10^{-3}$	0.561	0.567
t-statistics	-6.057	-0.581	-0.572
Action	Reject H_0	Fail to reject H_0	Fail to reject H_0

We observe that $p < 0.05$ at the surface level (indicating a significant increase in DO), but $p > 0.05$ at the middle and bottom levels (no statistically significant change).

BOD_5	Surface Level	Middle Level	Bottom Level

p-value	0.025	0.008	$<1\times10^{-3}$
t-statistics	2.247	2.648	4.440
Action	Reject H_0	Reject H_0	Reject H_0

For BOD_5, $p < 0.05$ at all levels, allowing us to reject H_0. We also obtain the type II error rate and the power:

	DO (Globally)	BOD5 (Globally)
Effect size (Welch-adjusted)	1.425	-3.091
Power ($1-\beta$)	~1.000	~1.000
Type II Error (β)	$<1\times10^{-3}$	$<1\times10^{-3}$

These results indicate high statistical power and negligible Type II error, reinforcing the reliability of the observed improvements. Based on these results, we summarize as follows:

- **Statistical Results:** Dissolved oxygen (DO) significantly increased at the surface level with a negative t-statistic, indicating $\mu_{(2020)} < \mu_{(2024)}$. No significant changes were observed at the middle or bottom levels. Biochemical oxygen demand (BOD_5) decreased significantly across all depths with positive t-statistics, indicating $\mu_{(2020)} > \mu_{(2024)}$. Both metrics show large effect sizes, near-perfect power, and minimal Type II error, reinforcing the conclusions.
- **Depth Patterns:** The most pronounced improvements occurred in surface water for both DO and BOD_5. Middle water showed significant BOD_5 reductions but no notable DO changes. Bottom water exhibited significant BOD_5 decreases, accompanied by mild, non-significant DO improvements.
- **Regional Patterns:** Most regions demonstrated increases in DO from 2020 to 2024, though significance varies. BOD_5 reductions were widespread but influenced by local pollution sources, with stronger improvements in areas like Victoria Harbour and Deep Bay.
- **Temporal Trends:** DO levels typically peaked in winter and dipped in summer. Monthly comparisons reveal consistently higher DO year-round in 2024 compared to 2020.

Conclusion

Based on a comprehensive analysis of DO and BOD_5 averages across three different depths and water zones in Hong Kong, we conclude that:

- There has been a significant overall improvement in marine water quality from 2020 to 2024, marked by reduced organic pollution at all depths.
- Dissolved oxygen levels increased in most regions, with the strongest gains in surface waters.
- These findings highlight the positive effects of environmental management measures, suggesting that sustained efforts could further enhance water quality.

Linear Regression

To further extend the conclusion above, we also attempted a linear regression test with the structure y = kx + b. We take other parameters in the database into consideration, and non-digital variables are converted into dummy variables. Using the OLS module in Python, we draw the linear regression fit diagram across multiple areas:

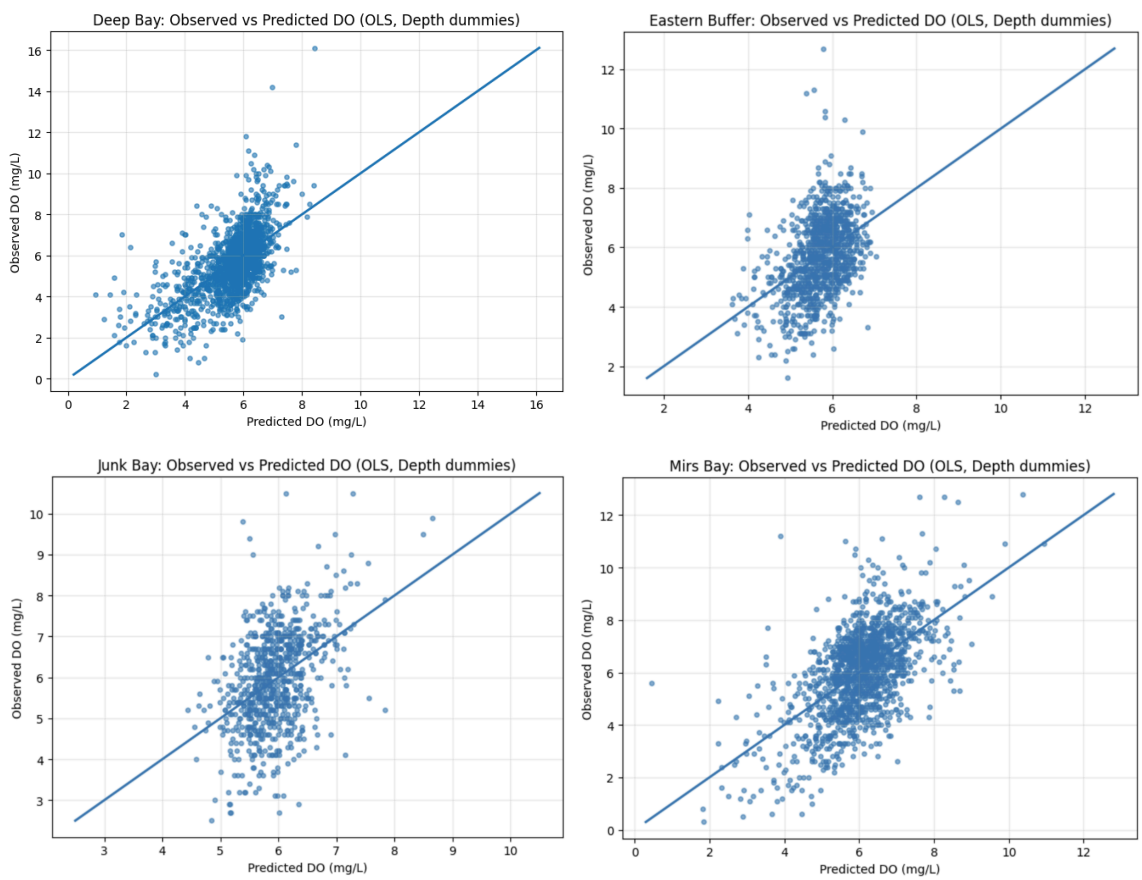

Fig. 8-11 Prediction fitness chart of predicted and observed DO in multiple areas (Not all areas shown)

Take Deep Bay as an example, the chart displays predicted dissolved oxygen (DO) values as a straight line, with predicted values shown on the X-axis and observed values on the Y-axis. The dots represent actual observations, and their proximity to the line indicates the accuracy of the

predictions. The model accounts for approximately 38% of the variation in DO, as indicated by an adjusted R^2 of about 0.382, which is not surprising, given the high viability of the natural environment. Most observations fall within the range of 4 to 7 mg/l and are closer to the predicted line, indicating that the model can explain most of the observations accurately. However, in the higher DO range (above 8 mg/l), some observations significantly exceed predictions, suggesting that the model has difficulties addressing abnormal values caused by certain extreme natural conditions such as algal blooms. Following this, we repeat the same procedures across all water zones, as demonstrated in Figures 8-11, where we observe the following:

- The model fails to capture extreme values resulting from emergencies or extreme natural weather, leaving abnormal dots on the plot. Better data cleaning in the preparation stage may address such issues.
- Model performance is strongest in Eastern Buffer, Tolo Harbour & Channel, and Deep Bay, and weaker in Victoria Harbour, Southern, and North Western, with a larger R-square. These differences reflect environmental variation: in clearer waters (e.g., Eastern Buffer, Mirs Bay), DO responds more directly to nutrients and solids, yielding stronger correlations. In contrast, urbanized zones with overlapping influences (e.g., wastewater, shipping) show weaker fits.
- Multicollinearity likely persists, which may result from high correlations between parameters such as Nitrogen metrics.

Discussion and Limitations

In summary, the linear regression analysis corroborates the findings of the t-test by demonstrating how factors such as regions, water depths, and other marine chemicals impact water quality, exhibiting moderate predictive capability in less urbanised areas. Together with the observed increases in DO (e.g, from 6.12 mg/L in 2020 to 6.40 mg/L in 2024 at surface levels) and decreases in BOD_5 (from 0.99 mg/L to 0.91 mg/L), these results imply genuine improvements due to environmental policies rather than natural variations alone. Consistent patterns across depths and months, such as winter peaks in DO, further support the efficacy of measures like enhanced sewage treatment in reducing organic pollution and improving marine ecological health.

This study has several limitations that must be addressed in future research. Firstly, the study focuses solely on dissolved oxygen (DO) and biochemical oxygen demand (BOD_5), overlooking important factors such as chemical contaminants and microorganisms that also affect water quality. Secondly, the dataset does not include data from 2021 to 2023, hindering our understanding of trends and the impact of events such as the global pandemic of 2020–21 on pollution levels. The regression models demonstrate moderate R^2 values and issues such as multicollinearity, suggesting that they may not fully capture the complexity of environmental

interactions. Sampling biases, including missing depth data and uneven station distribution, could also limit the generalisability of the study. Future research should incorporate more parameters, longer time series data, and advanced modelling techniques, such as multivariate regression or machine learning, to address these limitations and strengthen policy recommendations.

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